

DC014
Computer Science 1
Semester 1
Session 2022/2023
2 hours

DC014
Sains Komputer 1
Semester 1
Sesi 2022/2023
2 jam

No. Matrik Matric No.									

No. Kad Pengenalan Identity Card No.									

No. Tempat Duduk Seat No.			

(Isikan maklumat dengan lengkap)



KEMENTERIAN PENDIDIKAN MALAYSIA

KOLEJ MATRIKULASI SELANGOR
SELANGOR MATRICULATION COLLEGE

PRA PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
PRE MATRICULATION PROGRAMME SEMESTER EXAMINATION

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DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

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Untuk Kegunaan Pemeriksa For Examiner's Use			
No. Soalan Question No.	Markah Mark		
	Pemeriksa Examiner	KP / KKP	Markah Marks
1			6
2			16
3			6
4			4
5			18
6			20
JUMLAH TOTAL			70

Kertas soalan ini mengandungi 8 halaman bercetak.
This question paper consists of 8 printed pages.
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INSTRUCTIONS TO CANDIDATE :

Answer **all** the questions in the space provided.

- 1 Multimedia is very useful to human beings. It can be used to deliver information to the user through various ways.

a) Define Multimedia [2 marks]

b) There are five elements in multimedia. Animation is one of the multimedia elements. Give **two (2)** purposes of using animation. [2 marks]

c) List **two (2)** file formats for animation. [2 marks]

- 2 a) Multimedia design is a type of art that integrates multiple forms of multimedia. Differentiate between two-dimensional design, three-dimensional and four-dimensional design. [6 marks]

	Two-dimensional	Three-dimensional	Four-dimensional
Definition			
Usage			

- b) Authoring tools are used for designing interactivity and the user interface, presenting a project on screen and assembling diverse multimedia elements into a single, cohesive product. Name **three (3)** types of multimedia authoring tools. Give **one (1)** example for each type of authoring tools. [6 marks]

Authoring Tool	Example

- c) Compare **two (2)** differences between Web-based and CD-based in multimedia delivery. [4 marks]

Web-Based Multimedia Delivery	CD-based Multimedia Delivery

- 3 Database typically contain aggregations of data records or files , such as sales transactions, product catalogues and inventories and customer profiles.

- a) Define database management system (DBMS).

[2 marks]

b) State **one (1)** advantage of using database.

[2 marks]

c) Give **two (2)** components of database

[2 marks]

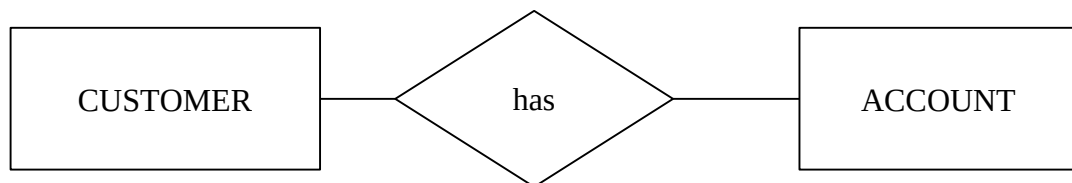
4 a) Data hierarchy includes **byte, table, bit, field** and **record**.
Rearrange the above data hierarchy from its low level to the highest level.

[1 mark]

b) Give the definition of **Entity Relationship Diagram (ERD)**.

[2 marks]

c) Based on the Entity Relationship Diagram below, identify the cardinality relationship between **CUSTOMER** and **ACCOUNT**.



[1 mark]

5 a) A network consists of two or more computers that are linked in order to share resources such as printers , exchange files, or allow electronic communications. The computers on a network may be linked through cables, telephone lines, radio waves, satellites, or infrared light beams.

Define network

[2 marks]

- b) The Global Positioning System (GPS) is a space-based radio-navigation system consisting of a constellation of satellites broadcasting navigation signals and a network of ground stations and satellite control stations used for monitoring and control.

i) Give **one(1)** of usage of network that enable to determine the current location other than GPS

[2 marks]

ii) Identify type of network media that allows GPS to receive and send data over Satellites

[2 marks]

- c) Given the following scenario:

An insurance company consisting of an owner, a business manager, an administrator and ten agents decides to implement a network. Everyone in the company has a desktop computer or a laptop and can share two printers, when agents need to print a document. All computers and devices are also connected to one server as a data source.

i) Identify **two(2)** type of network media used based on scenario above

[2 marks]

ii) State **two(2)** type of network device use to enable desktop and laptop connected through the network

[2 marks]

iii) What kind of network device is used to enable the printer to be shared.

[1 mark]

- d) State the name of network device base on their function.

Function	Network Device
It is a circuit board installed in a computer that provides a dedicated network connection to the computer	
connect multiple devices, such as computers, wireless access points, printers, and servers; on the same network within a building or campus.	
used to extend transmissions so that the signal can cover long distances	

[3 marks]

- e)

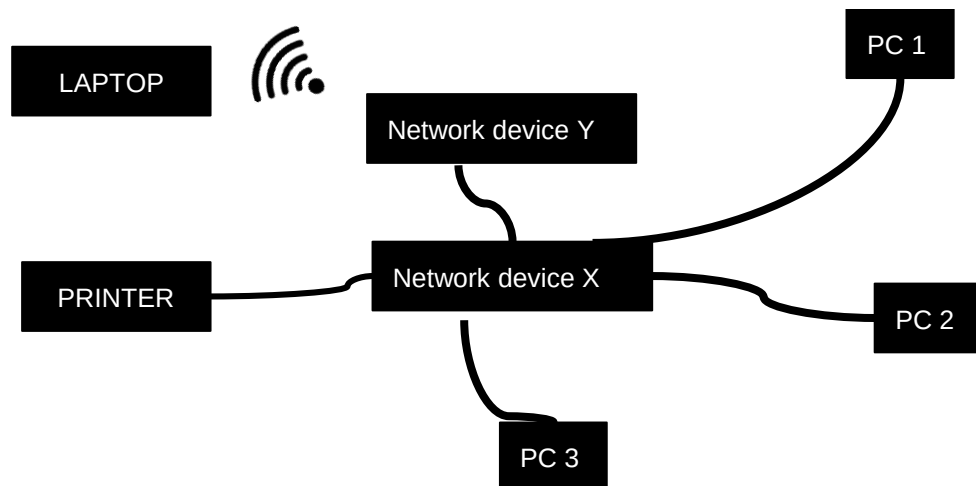


Figure 1: Office network

Based on **Figure 1**, answer the following question.

- i) State the name of network device X and network device Y

[2 marks]

network device X : _____

network device Y : _____

- ii) Identify example network media between :

[2 marks]

Network device Y to Laptop : _____

Network device X to Printer : _____

- 6 The Internet is a massive network of networks. It connects millions of computers together globally and it has become an integral part of our everyday life. A search tools is a program to locate the information on the Internet such as music files, image files, news articles, maps, people and videos

- a) State and explain **two (2)** types of search tools and give an example.

[6 marks]

Types of Search Tools	Explanation	Example

- b) When searching for information, there are a number of techniques to use that will help refine your search results. You can use these strategies when looking for information from the library databases or from a search engine such as Google.

- i. List **three (3)** search strategies can be used.

[3 marks]

- ii. Describe briefly **one (1)** of your answers.

[2 marks]

- c) A citation is a quotation from or reference to a book, paper or author, especially in a scholarly work. To avoid plagiarism, proper citation procedures should be used for all Internet content used in a paper, book, or on a Web site

- i. State **one (1)** example of citation format for web page article in online magazine.

[1 marks]

ii. Give **two (2)** importance to cite Internet resources.

[2 marks]

(a) _____

(b) _____

- d) Web activities include online communications, social networking/social media, e-commerce, online entertainment, online news, reference and information and also online education and writing.

Describe the following types of web activities.

[3 marks]

Types of web activity	Description
(a) e-commerce	
(b) online entertainment	
(c) instant messaging	

- e) Censorship is the suppression of speech, public communication or other information. This may be done on the basis that such material is considered objectionable, harmful, sensitive or "inconvenient".

i. Define censorship.

[1 marks]

- f) Internet of Things (IoT) refers to the collective network of connected devices and the technology that facilitates communication between devices and the cloud, as well as between the devices themselves.

Describe home automation.

[2 marks]

